

Final Project

2025-11-18

R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

Step 1: data loading

```
#install.packages("readxl")
library(readxl)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.1      v stringr   1.5.2
## v ggplot2    4.0.0      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(dplyr)
library(stringr)
library(tidyr)
library(ggplot2)
library(tibble)
library(knitr)
```

```
getwd()
```

```
## [1] "/Users/xuzhiwei/Desktop"
```

```
setwd("/Users/xuzhiwei/Documents/203A/Final Project")
```

```
data <- read_excel("PPMI_Curated_Data_Cut_Public_20251112.xlsx")
```

[1] 17252 179

tibble [17,252 x 179] (S3: tbl_df/tbl/data.frame)

```
## $ SITE : num [1:17252] 19 19 19 19 19 19 33 33 33 33 ...
## $ PATNO : chr [1:17252] "100001" "100001" "100001" "100001" ...
## $ COHORT : chr [1:17252] "1" "1" "1" "1" ...
## $ subgroup : chr [1:17252] "Sporadic PD" "Sporadic PD" "Sporadic PD" "Sporadic PD" ...
## $ enroll_phase : num [1:17252] 2 2 2 2 2 2 2 2 2 2 ...
## $ enroll_source : chr [1:17252] NA NA NA NA ...
## $ analytic_subgroup : chr [1:17252] NA NA NA NA ...
## $ HIQ_RBD : num [1:17252] NA NA NA NA NA NA NA NA NA ...
## $ study_status : chr [1:17252] "Active" "Active" "Active" "Active" ...
## $ NSD_Status : num [1:17252] 1 1 1 1 1 1 1 1 1 1 ...
## $ NSD_STAGE : chr [1:17252] "3" "3" "3" "3" ...
## $ PRIMDIAG : chr [1:17252] "1" "1" "1" "1" ...
## $ OTHNEURO : chr [1:17252] NA NA NA NA ...
## $ EVENT_ID : chr [1:17252] "BL" "V04" "V06" "V08" ...
## $ YEAR : num [1:17252] 0 1 2 3 4 5 0 1 2 3 ...
## $ visit_date : chr [1:17252] "10/2020" "11/2021" "11/2022" "11/2023" ...
## $ age : num [1:17252] 67.4 67.4 67.4 67.4 67.4 ...
## $ age_at_visit : num [1:17252] 67.4 68.5 69.5 70.5 71.3 ...
## $ SEX : num [1:17252] 1 1 1 1 1 1 1 1 1 1 ...
## $ EDUCYRS : num [1:17252] 16 16 16 16 16 16 18 18 18 18 ...
## $ race : num [1:17252] 1 1 1 1 1 1 1 1 1 1 ...
## $ HISPLAT : num [1:17252] 0 0 0 0 0 0 0 0 0 0 ...
## $ ASHKJEW : num [1:17252] 0 0 0 0 0 0 0 0 0 0 ...
## $ AFICBERB : num [1:17252] 0 0 0 0 0 0 0 0 0 0 ...
## $ BASQUE : num [1:17252] 0 0 0 0 0 0 0 0 0 0 ...
## $ fampd : num [1:17252] 2 2 2 2 2 2 2 2 2 2 ...
## $ fampd_bin : num [1:17252] 1 1 1 1 1 1 1 1 1 1 ...
## $ handed : num [1:17252] 1 1 1 1 1 1 1 1 1 1 ...
## $ howlive : num [1:17252] NA NA NA NA NA NA NA NA NA NA ...
## $ sex_orient : num [1:17252] NA NA NA NA NA NA NA NA NA NA ...
## $ BMI : num [1:17252] 24.6 24.9 23.6 24.1 23.5 ...
## $ ageddiag : num [1:17252] 66.3 66.3 66.3 66.3 66.3 ...
## $ ageonset : num [1:17252] 62.3 62.3 62.3 62.3 62.3 ...
## $ duration : num [1:17252] 12.7 12.7 12.7 12.7 12.7 ...
## $ duration_yrs : num [1:17252] 1.06 1.06 1.06 1.06 1.06 ...
## $ DOMSIDE : num [1:17252] 1 1 1 1 1 1 2 2 2 2 ...
## $ sym_tremor : num [1:17252] 0 0 0 0 0 0 1 1 1 1 ...
## $ sym_rigid : num [1:17252] 1 1 1 1 1 1 1 1 1 1 ...
## $ sym_brady : num [1:17252] 1 1 1 1 1 1 1 1 1 1 ...
## $ sym_posins : num [1:17252] 0 0 0 0 0 0 0 0 0 0 ...
## $ sym_other : num [1:17252] 0 0 0 0 0 0 0 0 0 0 ...
## $ sym_unknown : num [1:17252] 0 0 0 0 0 0 0 0 0 0 ...
## $ PDTRTMNT : chr [1:17252] "0" "0" "0" "0" ...
## $ LEDD : num [1:17252] 0 0 0 0 0 0 0 400 450 450 ...
## $ age_datscan : num [1:17252] 67.3 68.5 69.5 NA 71.3 ...
## $ age_LP : num [1:17252] 67.4 68.5 69.5 70.5 71.3 ...
## $ age_upsit : num [1:17252] 67.4 NA NA NA NA ...
## $ upsit : num [1:17252] 27 NA NA NA NA NA NA NA NA NA ...
## $ upsit_pctl : num [1:17252] 17.5 NA NA NA NA NA NA NA NA NA ...
## $ upsit_pctl15 : num [1:17252] 0 NA NA NA NA NA NA NA NA NA ...
## $ moca : num [1:17252] 26 26 25 26 25 25 26 27 26 28 ...
```

```
## $ bjplot : num [1:17252] 13 13 12 10 14 13 15 14 13 13 ...
## $ DVS_JLO_MSSA : num [1:17252] 13 13 12 9 14 13 16 14 13 13 ...
## $ DVS_JLO_MSSAE : num [1:17252] 12.16 12.16 11.06 7.76 13.26 ...
## $ clockdraw : num [1:17252] 5 7 7 5 4 7 6 7 7 7 ...
## $ DVT_CLCKDRAW : num [1:17252] 48 72.6 72.6 48 35.7 ...
## $ DVZ_CLCKDRAW : num [1:17252] -0.2 2.26 2.26 -0.2 -1.43 2.26 1.03 2.26 2.26 2.26 ...
## $ hvlt_discrimination : num [1:17252] 1 9 11 7 9 7 7 10 9 7 ...
## $ hvlt_immediatererecall : num [1:17252] 19 14 15 14 16 13 19 23 12 17 ...
## $ hvlt_retention : num [1:17252] 1 0.714 0.75 0.5 0.429 ...
## $ HVLTFPRL : num [1:17252] 1 2 0 0 2 2 2 1 1 3 ...
## $ HVLTRDLY : num [1:17252] 8 5 6 3 3 3 3 10 5 5 ...
## $ HVLTRC : num [1:17252] 2 11 11 7 11 9 9 11 10 10 ...
## $ DVT_TOTAL_RECALL : chr [1:17252] "33" "31" "33" "31" ...
## $ DVT_DELAYED_RECALL : chr [1:17252] "42" "36" "40" "29" ...
## $ DVT_RETENTION : chr [1:17252] "57" "42" "44" "31" ...
## $ DVT_RECOG_DISC_INDEX : chr [1:17252] "20" "42" "50" "31" ...
## $ lexical : num [1:17252] 32 38 29 40 44 46 28 29 34 42 ...
## $ DVT_FAS : num [1:17252] 41 48 41 48 52 56 34 37 41 49 ...
## $ DVS_FAS : num [1:17252] 8 10 8 10 11 12 7 8 9 11 ...
## $ lns : num [1:17252] 11 8 8 8 11 10 13 15 17 13 ...
## $ DVS_LNS : num [1:17252] 12 9 9 10 13 12 14 16 18 14 ...
## $ MODBNT : num [1:17252] 58 56 57 58 56 57 57 56 57 58 ...
## $ DVS_BNT : num [1:17252] 10 9 10 12 10 11 9 8 10 10 ...
## $ PCTL_BNT : chr [1:17252] "41 to 59" "29 to 40" "41 to 59" "72 to 81" ...
## $ SDMTOTAL : num [1:17252] 50 50 44 49 28 47 37 32 42 36 ...
## $ DVT_SDM : num [1:17252] 55.8 55.8 50 55 36 ...
## $ DVSD_SDM : num [1:17252] 0.583 0.583 0 0.5 -1.4 ...
## $ TMT_A : num [1:17252] 43 40 43 37 40 31 28 33 37 42 ...
## $ TMT_B : num [1:17252] 87 59 73 97 106 72 60 70 58 97 ...
## $ DVZ_TMTA : num [1:17252] -1.37 -0.92 -1.37 0.22 0.01 0.63 0.37 -0.13 -0.82 -1.53 ...
## $ DVZ_TMTB : num [1:17252] -2.14 0.87 -0.63 -0.45 -0.82 0.59 0.42 -0.06 0.35 -1.74 ...
## $ VLTANIM : num [1:17252] 19 22 23 16 20 17 19 20 14 23 ...
## $ DVT_SFTANIM : num [1:17252] 49 53 57 43 51 47 45 45 35 53 ...
## $ DVS_SFTANIM : num [1:17252] 10 11 12 8 10 9 10 10 7 12 ...
## $ MCI_testscores : num [1:17252] 1 0 0 1 0 1 1 0 1 1 ...
## $ cogstate : num [1:17252] 1 1 1 1 1 1 1 1 1 1 ...
## $ COG_COMPOSITE_INT : num [1:17252] -0.39 -0.891 -0.929 -1.491 -1.106 ...
## $ COG_COMPOSITE_EXT : num [1:17252] -0.105 -0.338 -0.33 -0.824 -0.469 ...
## $ MSEADLG : num [1:17252] 90 95 95 95 90 95 95 95 95 90 ...
## $ quip : num [1:17252] 0 0 1 0 0 0 0 0 0 0 ...
## $ quip_any : num [1:17252] 0 0 1 0 0 0 0 0 0 0 ...
## $ quip_gamble : num [1:17252] 0 0 0 0 0 0 0 0 0 0 ...
## $ quip_sex : num [1:17252] 0 0 0 0 0 0 0 0 0 0 ...
## $ quip_buy : num [1:17252] 0 0 0 0 0 0 0 0 0 0 ...
## $ quip_eat : num [1:17252] 0 0 1 0 0 0 0 0 0 0 ...
## $ quip_hobby : num [1:17252] 0 0 0 0 0 0 0 0 0 0 ...
## $ quip_pund : num [1:17252] 0 0 0 0 0 0 0 0 0 0 ...
## $ quip_walk : num [1:17252] 0 0 0 0 0 0 0 0 0 0 ...
## [list output truncated]
```

Step 2: Data Cleaning & Filtering

```
clean_data <- data %>%
  # Keep only Baseline (BL) data
  filter(EVENT_ID == "BL") %>%

  # 1 = PD, 2 = Healthy Control, 4 = Prodromal
  filter(COHORT %in% c(1, 2, 4)) %>%

  # Create Labels
  mutate(
    Group = case_when(
      COHORT == 1 ~ "PD",
      COHORT == 2 ~ "Healthy Control",
      COHORT == 4 ~ "Prodromal"
    ),
    # Set factor levels order: Healthy -> Prodromal -> PD
    Group = factor(Group, levels = c("Healthy Control", "Prodromal", "PD"))
  ) %>%

  # Select Variables
  select(
    PATNO,          # Patient ID
    Group,          # Cohort Group
    age,            # Age
    SEX,            # Sex

    # Early Risk Factors (Non-Motor)
    upsit,          # Smell (Lower = Worse)
    rem,            # Sleep / RBD (Higher = Worse)
    scopa,          # Autonomic (Higher = Worse)

    # Clinical Symptoms
    updrs3_score,   # Motor Symptoms (Higher = Worse)
    moca,           # Cognition (Lower = Worse)

    # Biomarkers & Genetics
    subgroup,       # Genetic Subgroups
    MIA_PUTAMEN_L   # Brain Dopamine (DaTscan Left Putamen)
  )

# 3. Check Data
summary(clean_data %>% select(age, upsit, rem, updrs3_score, moca))
```

```
##      age      upsit      rem      updrs3_score
## Min.   :26.44  Min.    : 0.00  Min.    : 0.000  Min.     : 0.00
## 1st Qu.:61.09  1st Qu. :17.00  1st Qu. : 2.000  1st Qu.  : 1.00
## Median :65.98  Median  :24.00  Median  : 4.000  Median   : 6.00
## Mean   :65.23  Mean    :24.09  Mean    : 4.843  Mean     :10.34
## 3rd Qu.:70.80  3rd Qu. :31.00  3rd Qu. : 8.000  3rd Qu.  :17.00
## Max.   :91.41  Max.    :40.00  Max.    :13.000  Max.     :71.00
##      NA's      NA's      NA's      NA's
##      moca
```

```
## Min.      :11.00
## 1st Qu.:25.00
## Median :27.00
## Mean    :26.73
## 3rd Qu.:29.00
## Max.     :30.00
## NA's     :21
```

Data Collection Flow Chart

```
n_raw <- nrow(data) # Original number of rows
n_baseline <- data %>% filter(EVENT_ID == "BL") %>% nrow()
n_final <- clean_data %>% nrow()

box_data <- tibble(
  x = c(0.5, 0.5, 0.5),
  y = c(0.9, 0.5, 0.1),
  label = c(
    paste0("Raw Data\n(N = ", n_raw, ")"),
    paste0("Baseline Visits Only\n(Event_ID = 'BL')\n(N = ", n_baseline, ")"),
    paste0("Final Analytical Cohort\n(Healthy, PD, Prodromal)\n(N = ", n_final, ")")
  )
)

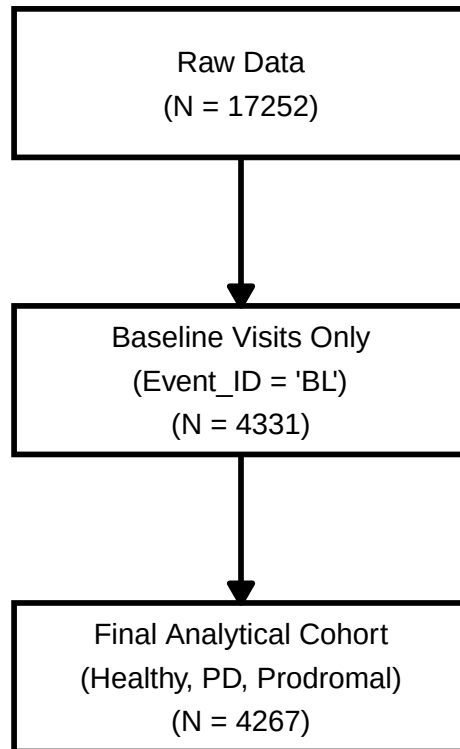
arrow_data <- tibble(
  x = c(0.5, 0.5),
  xend = c(0.5, 0.5),
  y = c(0.8, 0.4),
  yend = c(0.6, 0.2)
)

ggplot() +
  # Draw Boxes
  geom_rect(data = box_data,
    aes(xmin = x - 0.2, xmax = x + 0.2, ymin = y - 0.1, ymax = y + 0.1),
    fill = "white", color = "black", size = 1) +
  # Add Text
  geom_text(data = box_data, aes(x = x, y = y, label = label), size = 4) +
  # Add Arrows
  geom_segment(data = arrow_data,
    aes(x = x, y = y, xend = xend, yend = yend),
    arrow = arrow(length = unit(0.3, "cm"), type = "closed"),
    size = 1) +
  theme_void() +
  ylim(0, 1) +
  xlim(0, 1) +
  labs(title = "Figure 1: Data Selection Flow Chart")
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
```

generated.

Figure 1: Data Selection Flow Chart



Descriptive Statistics of Study Measurements

```
table_data <- clean_data %>%
  select(Group,
    `Age` = age,
    `Smell (UPSIT)` = upsit,
    `Sleep (RBD)` = rem,
    `Autonomic (SCOPA)` = scopa,
    `Motor (UPDRS III)` = updrs3_score,
    `Cognition (MoCA)` = moca,
    `Dopamine (DaTscan)` = MIA_PUTAMEN_L) %>%
  pivot_longer(cols = -Group, names_to = "Measurement", values_to = "Value")

summary_table <- table_data %>%
  group_by(Group, Measurement) %>%
  summarise(
    N = sum(!is.na(Value)),
    Mean = round(mean(Value, na.rm = TRUE), 2),
    SD = round(sd(Value, na.rm = TRUE), 2),
    Median = round(median(Value, na.rm = TRUE), 2),
    Min = round(min(Value, na.rm = TRUE), 2),
    Max = round(max(Value, na.rm = TRUE), 2),
    .groups = 'drop'
  ) %>%
```

```
arrange(Group, Measurement)
```

```
## Warning: There were 2 warnings in 'summarise()'.
## The first warning was:
## i In argument: 'Min = round(min(Value, na.rm = TRUE), 2)'.
## i In group 11: 'Group = Prodromal' 'Measurement = "Dopamine (DaTscan)"'.
## Caused by warning in 'min()':
## ! no non-missing arguments to min; returning Inf
## i Run 'dplyr::last_dplyr_warnings()' to see the 1 remaining warning.
```

```
kable(summary_table, caption = "Table 1: Descriptive Statistics of Study Measurements")
```

Table 1: Table 1: Descriptive Statistics of Study Measurements

Group	Measurement	N	Mean	SD	Median	Min	Max
Healthy Control	Age	336	62.12	11.37	63.77	30.41	85.34
Healthy Control	Autonomic (SCOPA)	334	6.12	4.17	5.00	0.00	29.00
Healthy Control	Cognition (MoCA)	336	27.83	1.70	28.00	20.00	30.00
Healthy Control	Dopamine (DaTscan)	322	1.55	0.31	1.50	0.90	2.60
Healthy Control	Motor (UPDRS III)	332	1.49	2.31	0.00	0.00	13.00
Healthy Control	Sleep (RBD)	336	2.55	2.39	2.00	0.00	13.00
Healthy Control	Smell (UPSIT)	332	33.44	5.51	35.00	10.00	40.00
Prodromal	Age	2456	67.11	6.78	67.06	26.44	91.41
Prodromal	Autonomic (SCOPA)	2446	10.52	6.74	9.00	0.00	43.00
Prodromal	Cognition (MoCA)	2440	26.59	2.51	27.00	11.00	30.00
Prodromal	Dopamine (DaTscan)	0	NaN	NA	NA	Inf	-Inf
Prodromal	Motor (UPDRS III)	2408	4.61	5.76	3.00	0.00	48.00
Prodromal	Sleep (RBD)	2442	5.60	4.09	5.00	0.00	13.00
Prodromal	Smell (UPSIT)	2441	23.59	8.24	23.00	2.00	40.00
PD	Age	1475	62.83	9.84	63.89	28.35	85.25
PD	Autonomic (SCOPA)	1452	10.81	7.06	10.00	0.00	51.00
PD	Cognition (MoCA)	1470	26.72	2.65	27.00	11.00	30.00
PD	Dopamine (DaTscan)	1396	0.71	0.30	0.67	0.03	2.18
PD	Motor (UPDRS III)	1382	22.45	10.12	21.00	2.00	71.00
PD	Sleep (RBD)	1467	4.11	2.96	3.00	0.00	13.00
PD	Smell (UPSIT)	1431	22.76	8.17	23.00	0.00	40.00

Step 3: Visualization

```
plot_data <- clean_data %>%
  select(Group, upsit, rem, scopa, updrs3_score, moca) %>%
  pivot_longer(
    cols = -Group,
    names_to = "Metric",
    values_to = "Value"
  )

metric_labels <- c(
  "upsit" = "Smell (UPSIT) - Lower is Worse",
```

```

"rem" = "Sleep (RBD Score) - Higher is Worse",
"scopa" = "Autonomic (SCOPA) - Higher is Worse",
"updrs3_score" = "Motor (UPDRS III) - Higher is Worse",
"moca" = "Cognition (MoCA) - Lower is Worse"
)

ggplot(plot_data, aes(x = Group, y = Value, fill = Group)) +
  # Boxplot
  geom_boxplot(alpha = 0.7, outlier.shape = NA) +
  # Jitter points for detail
  geom_jitter(width = 0.1, alpha = 0.1, size = 0.05) +

  # Facet wrap to show 5 plots
  facet_wrap(~Metric, scales = "free_y", labeller = as_labeller(metric_labels)) +

  theme_light() +
  scale_fill_brewer(palette = "Set2") +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "top"
  ) +
  labs(
    title = "Comparison of Early PD Risk Factors",
    subtitle = "Comparing Healthy Control vs. Prodromal vs. PD",
    x = "",
    y = "Score",
    fill = "Cohort"
  )

```

```

## Warning: Removed 286 rows containing non-finite outside the scale range
## ('stat_boxplot()').

```

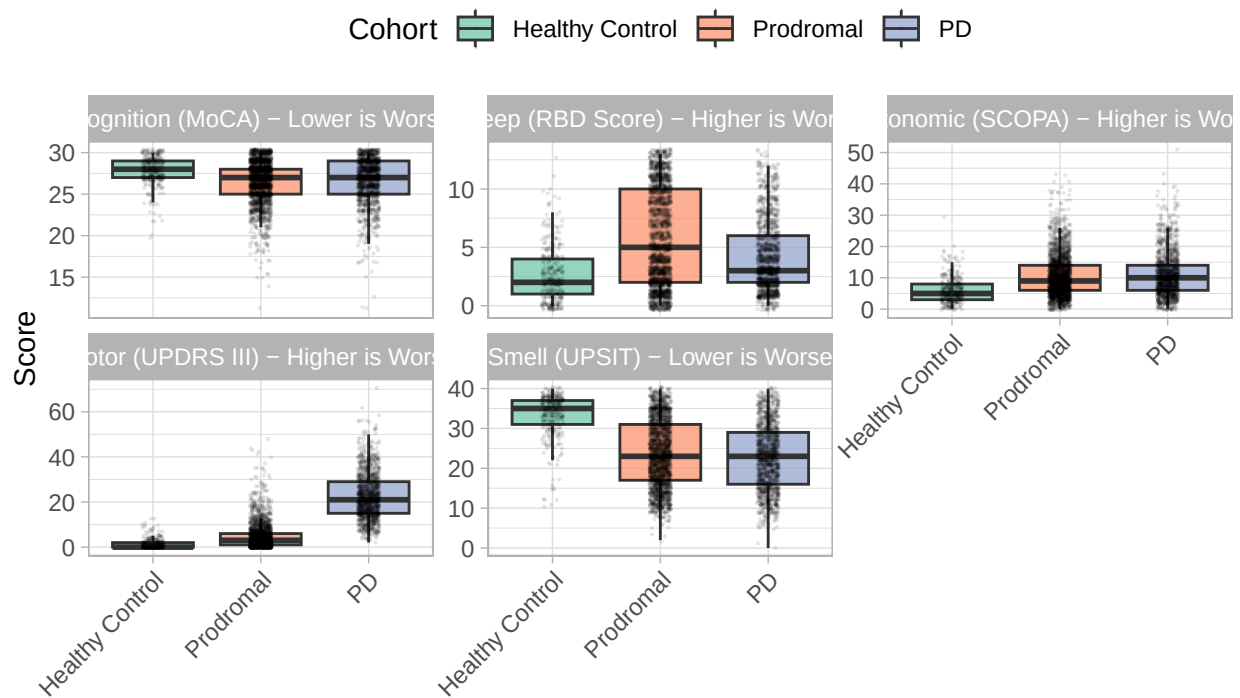
```

## Warning: Removed 286 rows containing missing values or values outside the scale range
## ('geom_point()').

```


Comparison of Early PD Risk Factors

Comparing Healthy Control vs. Prodromal vs. PD



```
# 1. Categorize Genetics
pd_genetic_data <- clean_data %>%
  filter(Group == "PD") %>% # Select only PD patients
  mutate(
    Genetic_Status = case_when(
      grepl("LRRK2", subgroup) ~ "LRRK2",
      grepl("GBA", subgroup) ~ "GBA",
      grepl("SNCA", subgroup) ~ "SNCA",
      TRUE ~ "Sporadic" # Default to Sporadic if no mutation found
    )
  )

# 2. Plot A: Genetics vs. Smell (UPSIT)
p1 <- ggplot(pd_genetic_data, aes(x = Genetic_Status, y = upsit, fill = Genetic_Status)) +
  geom_boxplot(alpha = 0.7, outlier.shape = NA) +
  geom_jitter(width = 0.2, alpha = 0.3, size = 1) +
  scale_fill_viridis_d() +
  theme_light() +
  labs(
    title = "Effect of Genotype on Smell (PD Only)",
    subtitle = "Note: LRRK2 patients often have better smell (higher scores)",
    x = "Genetic Subtype",
    y = "UPSIT Score (Smell)"
  ) + theme(legend.position = "none")

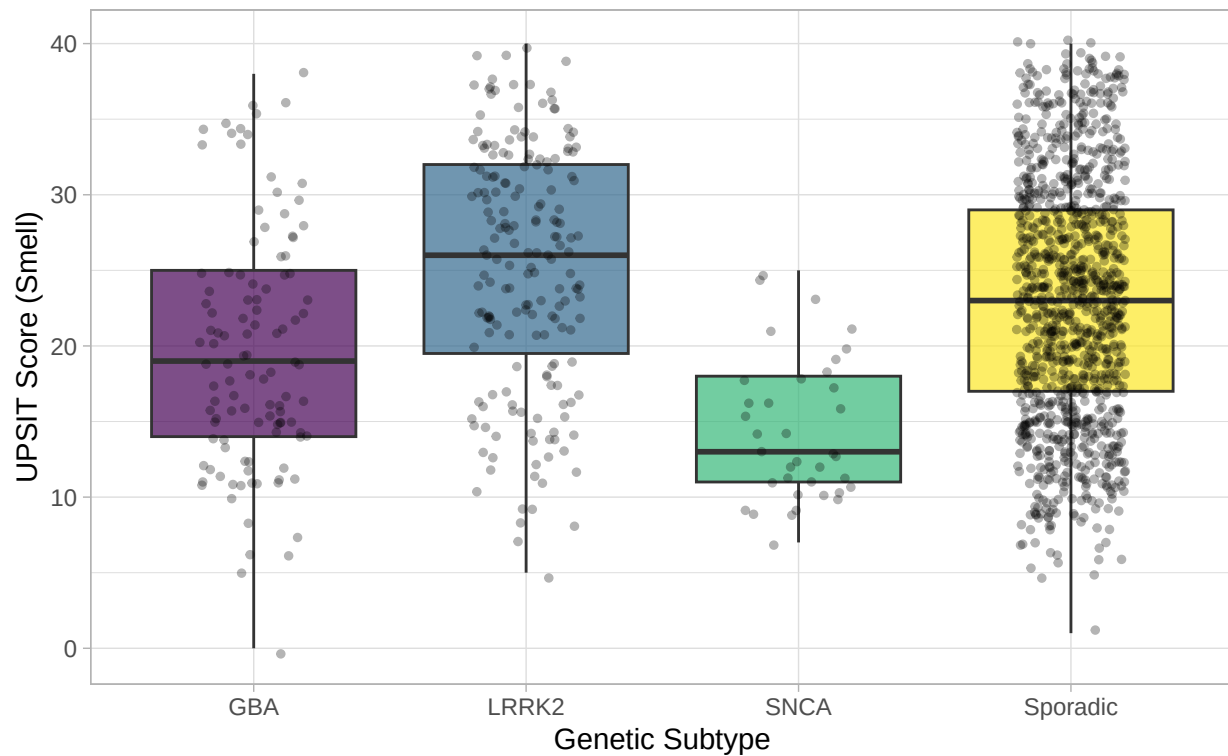
print(p1)
```

```
## Warning: Removed 44 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Warning: Removed 44 rows containing missing values or values outside the scale range
## ('geom_point()').
```

Effect of Genotype on Smell (PD Only)

Note: LRRK2 patients often have better smell (higher scores)



```
# 3. Plot B: Genetics vs. Cognition (MoCA)
p2 <- ggplot(pd_genetic_data, aes(x = Genetic_Status, y = moca, fill = Genetic_Status)) +
  geom_boxplot(alpha = 0.7, outlier.shape = NA) +
  geom_jitter(width = 0.2, alpha = 0.3, size = 1) +
  scale_fill_viridis_d(option = "magma") +
  theme_light() +
  labs(
    title = "Effect of Genotype on Cognition (PD Only)",
    subtitle = "Note: GBA patients may have lower cognitive scores",
    x = "Genetic Subtype",
    y = "MoCA Score (Cognition)"
  ) + theme(legend.position = "none")

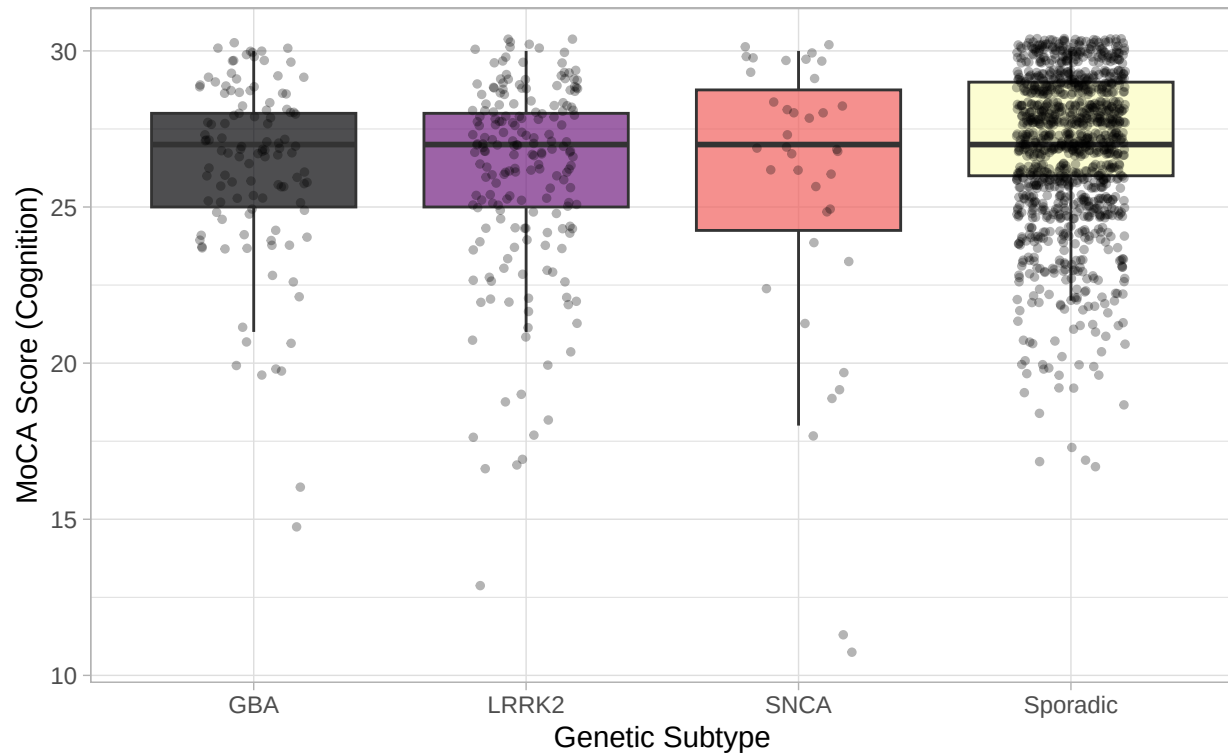
print(p2)
```

```
## Warning: Removed 5 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

```
## Warning: Removed 5 rows containing missing values or values outside the scale range
## ('geom_point()').
```

Effect of Genotype on Cognition (PD Only)

Note: GBA patients may have lower cognitive scores



```
# 1. Correlation: Smell vs. Brain Dopamine
cor_smell <- cor.test(clean_data$upsit, clean_data$MIA_PUTAMEN_L, use = "complete.obs")
```

```
# Plot 1: Smell vs Dopamine
```

```
p3 <- ggplot(clean_data, aes(x = upsit, y = MIA_PUTAMEN_L)) +
  geom_point(aes(color = Group), alpha = 0.6, size = 1.5) +
  geom_smooth(method = "lm", color = "black", se = TRUE) +
  theme_light() +
  scale_color_brewer(palette = "Set2") +
  labs(
    title = "Biological Evidence: Smell Loss vs. Brain Dopamine",
    subtitle = paste("Correlation (r) =", round(cor_smell$estimate, 3),
      "- Worse Smell (Left) correlates with Lower Dopamine (Bottom)",
    x = "UPSIT Score (Smell)",
    y = "Putamen DaTscan (Dopamine Level)",
    color = "Cohort"
  )
```

```
print(p3)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

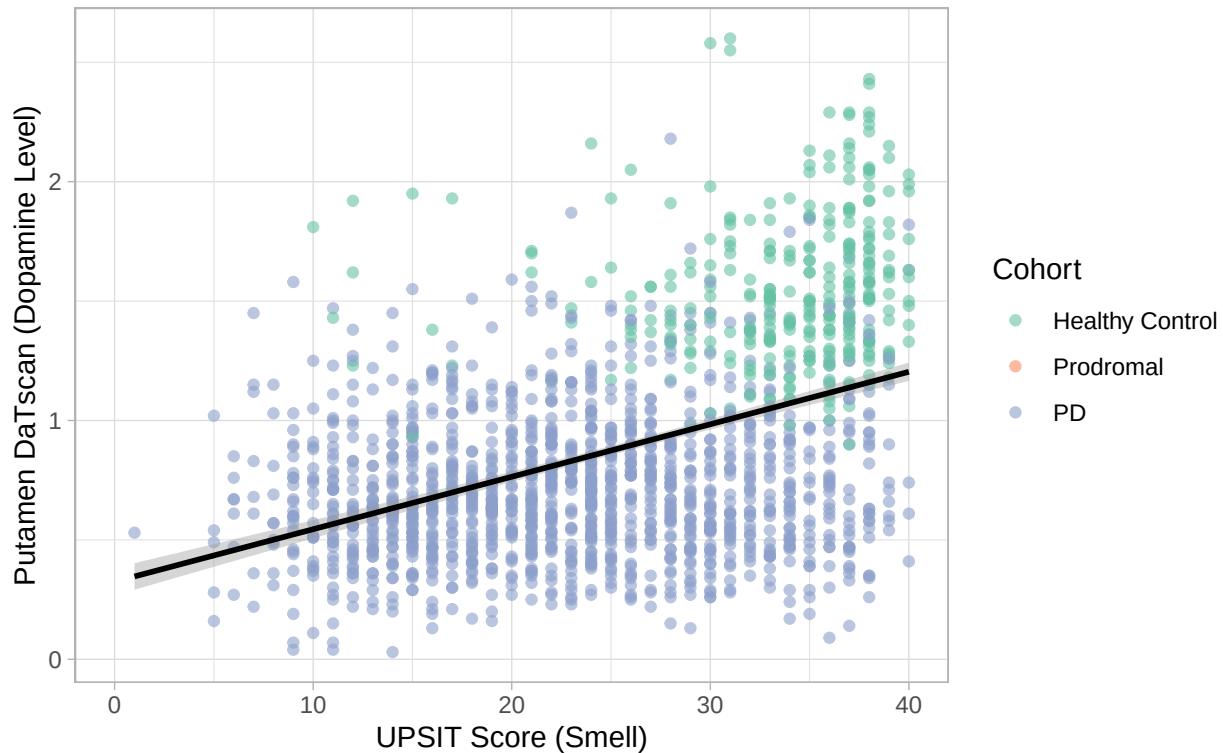
```
## Warning: Removed 2590 rows containing non-finite outside the scale range
## ('stat_smooth()').
```

```
## Warning: Removed 2590 rows containing missing values or values outside the scale range
```

```
## ('geom_point()').
```

Biological Evidence: Smell Loss vs. Brain Dopamine

Correlation (r) = 0.434 – Worse Smell (Left) correlates with Lower Dopamine (Bottom)



```
# 2. Correlation: Motor Symptoms vs. Brain Dopamine
cor_motor <- cor.test(clean_data$updrs3_score, clean_data$MIA_PUTAMEN_L, use = "complete.obs")

# Plot 2: Motor vs Dopamine
p4 <- ggplot(clean_data, aes(x = MIA_PUTAMEN_L, y = updrs3_score)) +
  geom_point(aes(color = Group), alpha = 0.6) +
  geom_smooth(method = "lm", color = "black") +
  theme_light() +
  scale_color_brewer(palette = "Set1") +
  labs(
    title = "Biological Evidence: Dopamine Loss vs. Motor Symptoms",
    subtitle = paste("Correlation (r) =", round(cor_motor$estimate, 3),
      "(Negative correlation: Lower Dopamine = Higher/Worse Motor Score)"),
    x = "Putamen DaTscan (Dopamine Level)",
    y = "MDS-UPDRS III Score (Motor Symptoms)",
    color = "Cohort"
  )

print(p4)
```

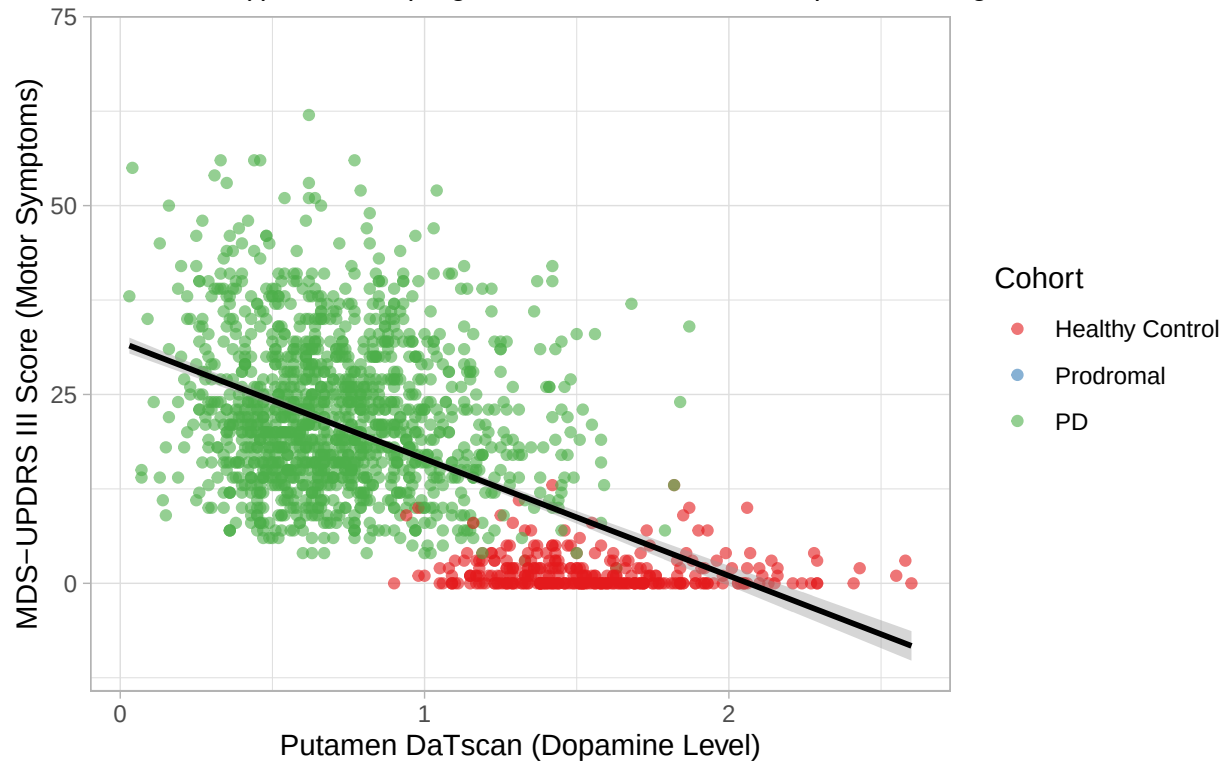
```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 2618 rows containing non-finite outside the scale range
## ('stat_smooth()').
```

```
## Warning: Removed 2618 rows containing missing values or values outside the scale range
## ('geom_point()').
```

Biological Evidence: Dopamine Loss vs. Motor Symptoms

Correlation (r) = -0.562 (Negative correlation: Lower Dopamine = Higher/Worse Motor Sc



```
reg_data <- clean_data %>%
  drop_na(upsit, rem, scopa, MIA_PUTAMEN_L)

logit_data <- reg_data %>%
  filter(Group %in% c("PD", "Healthy Control")) %>%
  mutate(
    # Create Target Variable: 1 = PD (Disease), 0 = Healthy
    is_PD = ifelse(Group == "PD", 1, 0)
  )

model_logit <- glm(is_PD ~ upsit + rem + scopa + age + SEX,
  data = logit_data,
  family = binomial)

summary(model_logit)
```

```
##
## Call:
## glm(formula = is_PD ~ upsit + rem + scopa + age + SEX, family = binomial,
##      data = logit_data)
##
## Coefficients:
```

```
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept)  9.061298   0.738238  12.274 < 2e-16 ***
## upsit       -0.230131   0.014872 -15.474 < 2e-16 ***
## rem         0.095949   0.034831   2.755 0.00587 **
## scopas      0.142303   0.019747   7.206 5.75e-13 ***
## age        -0.036723   0.007948  -4.620 3.83e-06 ***
## SEX        -0.095346   0.162363  -0.587 0.55704
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 1613.5  on 1653  degrees of freedom
## Residual deviance: 1003.3  on 1648  degrees of freedom
## AIC: 1015.3
##
## Number of Fisher Scoring iterations: 6
```

```
exp(coef(model_logit))
```

```
## (Intercept)      upsit      rem      scopas      age      SEX
## 8615.3282131    0.7944296    1.1007030    1.1529265    0.9639428    0.9090587
```

```
model_linear <- lm(MIA_PUTAMEN_L ~ upsit + rem + scopas + age,
  data = reg_data)
```

```
print("--- Brain Dopamine Prediction Model ---")
```

```
## [1] "--- Brain Dopamine Prediction Model ---"
```

```
summary(model_linear)
```

```
##
## Call:
## lm(formula = MIA_PUTAMEN_L ~ upsit + rem + scopas + age, data = reg_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.06196 -0.26655 -0.03483  0.22881  1.52721
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.1853886  0.0741059   2.502  0.01246 *
## upsit       0.0207798  0.0011387  18.249 < 2e-16 ***
## rem        -0.0115429  0.0035916  -3.214  0.00133 **
## scopas     -0.0100395  0.0015839  -6.339 2.99e-10 ***
## age         0.0050121  0.0009883   5.072 4.39e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3909 on 1649 degrees of freedom
## Multiple R-squared:  0.2318, Adjusted R-squared:  0.23
## F-statistic: 124.4 on 4 and 1649 DF, p-value: < 2.2e-16
```

```
library(broom)

tidy_logit <- tidy(model_logit, exponentiate = TRUE, conf.int = TRUE) %>%
  filter(term != "(Intercept)") # Remove Intercept

ggplot(tidy_logit, aes(x = estimate, y = term, xmin = conf.low, xmax = conf.high)) +
  geom_point(color = "red", size = 3) +
  geom_errorbarh(height = 0.2) +
  geom_vline(xintercept = 1, linetype = "dashed") + # 1 represents no effect
  labs(
    title = "PD Risk Factor Odds Ratios",
    subtitle = "Left of 1 = Protective Factor, Right of 1 = Risk Factor",
    x = "Odds Ratio",
    y = "Risk Factor"
  ) +
  theme_minimal()
```

```
## Warning: 'geom_erorbarh()' was deprecated in ggplot2 4.0.0.
## i Please use the 'orientation' argument of 'geom_errorbar()' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
## 'height' was translated to 'width'.
```

PD Risk Factor Odds Ratios

Left of 1 = Protective Factor, Right of 1 = Risk Factor

